Abstract

Advances in artificial intelligence (AI) have enabled the wide use of personalized assistance for cognitive tasks, which enhanced individual performance on various fields. However, the impact of AI assistance on collective intelligence remains poorly understood. AI assistance may undermine the emergence of collective intelligence by reducing the individual diversity or introducing structured biases into a population. Here, we develop a human-AI collective prediction model mediated by social learning based on incentive structure by extending the existing works on collective intelligence. Our model uncovers that emergence of collective intelligence is depend on how human and AI interact. When AI works as if it is omniscient and gives assistance regardless of users' interests, collective intelligence will be promoted. However, when AI works as a chat-bot and gives assistance only based on users' interests, collective intelligence could be hindered due to diversity loss and overconfidence on AI. We showed that, to promote collective intelligence, penalizing or incentivizing AI usage is not reliable. Instead, elaborating more diversity-aware incentives could be the key to promote and maintain the collective intelligence.

Introduction

The emergence of large language models (LLMs) has changed the way individuals engage in cognitive tasks. As AI services become increasingly popular, individuals can obtain personalized assistance from a shared source of intelligence. Empirical research has shown that AI assistance can improve individual performance in various domains [1], including economic forecasting [2], medical decision-making [3], and creative writing [4]. However, its effects on collective performance remain poorly understood. Some researchers suggested that AI can promote the emergence of collective intelligence by enhancing interdisciplinary interaction, opinion aggregation, and idea generation [5]. Some have even argued that human-AI collective intelligence could overcome the limitations of human-only collective intelligence [6]. Conversely, there are large concerns that AI could undermine collective intelligence by reducing individual diversity, propagating illusions of consensus, or generating false information [5]. Prior research has shown that AI assistance could hinder the wisdom of crowds by reducing unique human knowledge [7].

To date, existing studies have mainly focused on the static environment without dynamic social learning among individuals [1, 7]. Because social learning is pervasive in human populations, a modeling approach that incorporates social learning is essential for gaining a comprehensive understanding of the long-term effects of AI. In research on how social learning gives rise to collective intelligence, incentive structures have been recognized as important drivers [8-10]. When an appropriate incentive structure is given, collective intelligence can emerge and be maintained from the randomized population [8, 9]. While the use of AI assistance has also become part of individual strategy, social learning based on the incentive structure would affect the interaction between humans and AI, thereby influencing the emergence of collective intelligence.

Because of the growing impact of AI services, different societies have attempted to modify incentive structures to improve collective performance. In some domains, such as education and academia, the use of AI is treated as a form of deception and is penalized through regulations [11]. In contrast, in engineering and industry, the use of AI is often regarded as essential for productivity, and in some cases, is directly incentivized. It is therefore crucial to understand the consequences of directly penalizing or incentivizing AI usage for collective intelligence. Moreover, finding a robust incentive structure that can promote collective intelligence in the presence of AI could provide insight into how societies can prevent risks that could be caused by AI.

Here, we developed a human-AI collective prediction model with social learning, based on previous modeling approaches. We examined two extreme forms of human-AI interaction. First is “AI knows all” model, in which a single AI model covers all the interests and provides assistance to human users regardless of their interest. The second is “AI answers question” model, in which an AI model is composed of multiple expert models that give assistance only about each user’s interests.

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